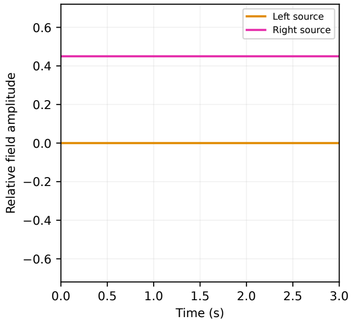
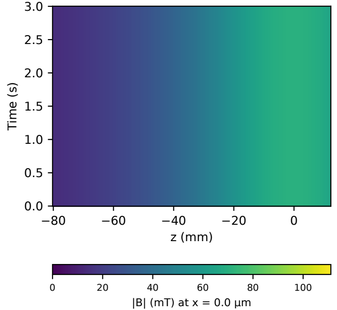
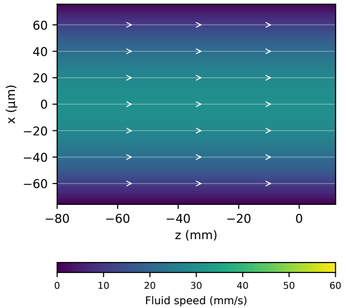
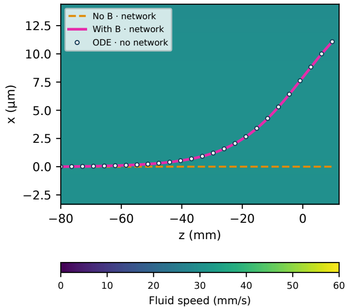
# TRAINED MASTER10 | Constant right source

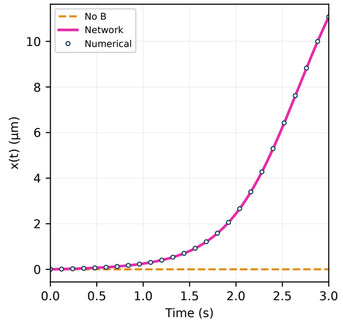
Original case · $x_0$=0.00 µm, $z_0$=-80.0 mm · Umean=15 mm/s, η=3.277 mPa·s · T=3 s

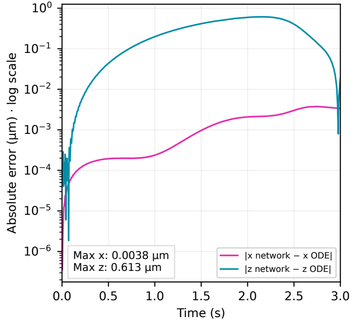


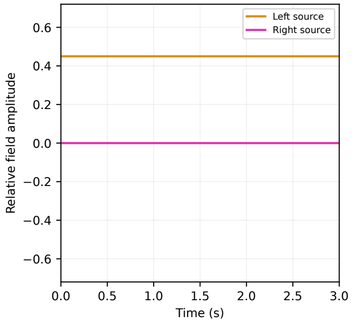

Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Original straight-channel geometry and overdamped equation. No independent launch-velocity input. No-B paths remain straight. Spatial axes use different units.

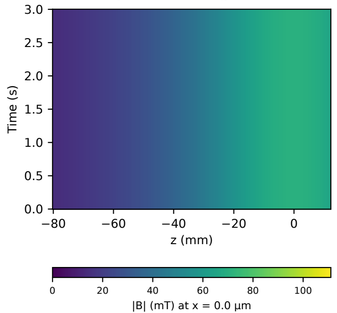
# TRAINED MASTER10 | Constant left source

Original case · $x_0$=0.00 µm, $z_0$=-80.0 mm · Umean=15 mm/s, η=3.277 mPa·s · T=3 s

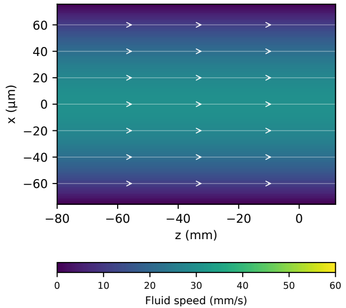
## 01 Magnetic input



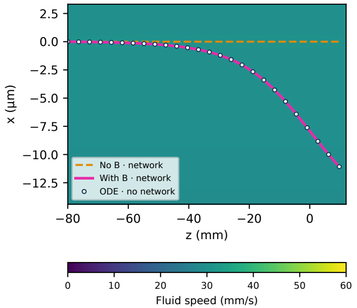

## 02 Space-time magnetic field



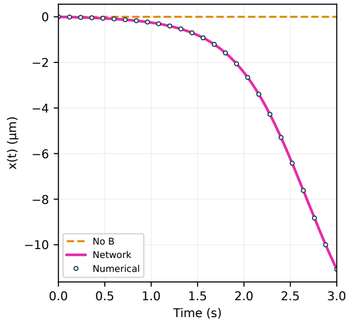

## 03 Trained-domain fluid flow



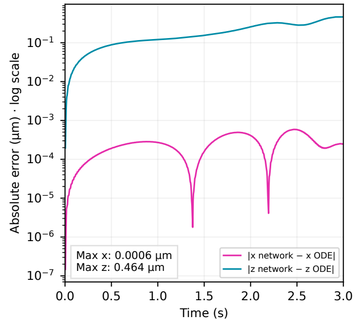

## 04 Network / numerical paths




## 05 Transverse response




## 06 Actual prediction error



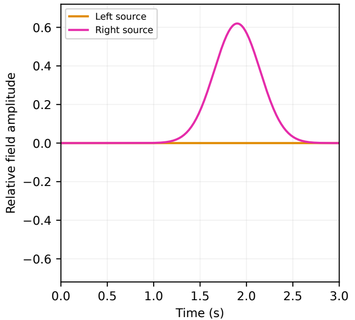

Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Original straight-channel geometry and overdamped equation. No independent launch-velocity input. No-B paths remain straight. Spatial axes use different units.

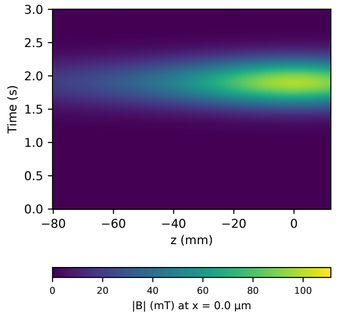
# TRAINED MASTER10 | Gaussian pulse on the right

Original case · $x_0$=0.00 µm, $z_0$=-80.0 mm · Umean=15 mm/s, η=3.277 mPa·s · T=3 s

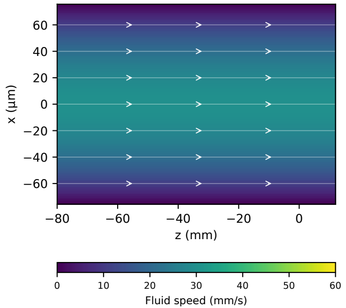
## 01 Magnetic input



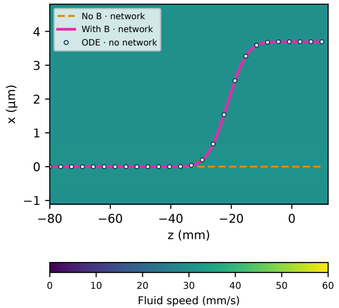

## 02 Space–time magnetic field



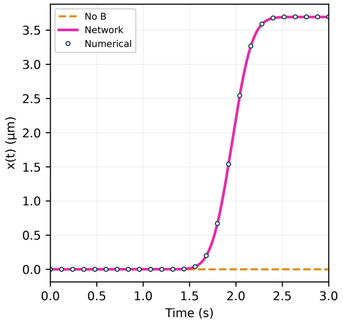

## 03 Trained-domain fluid flow



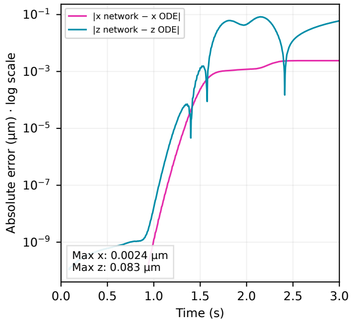

## 04 Network / numerical paths




## 05 Transverse response




## 06 Actual prediction error



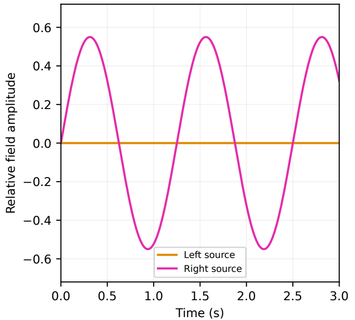

Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Original straight-channel geometry and overdamped equation. No independent launch-velocity input. No-B paths remain straight. Spatial axes use different units.

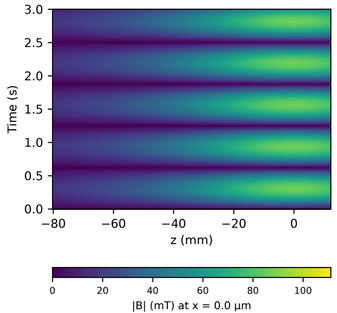
# TRAINED MASTER10 | Sinusoidal right source

Original case · $x_0$=0.00 µm, $z_0$=-80.0 mm · Umean=15 mm/s, η=3.277 mPa·s · T=3 s

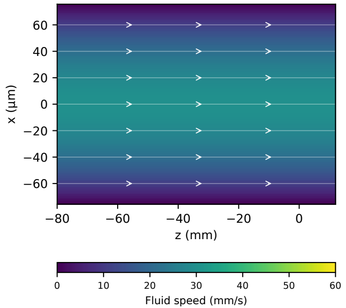
## 01 Magnetic input



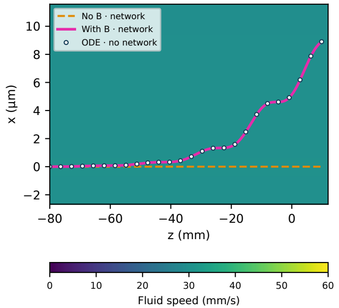

## 02 Space-time magnetic field



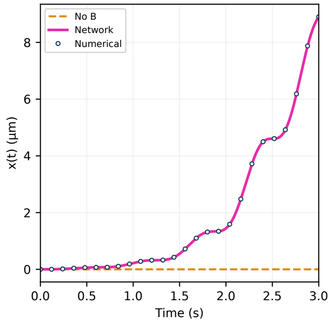

## 03 Trained-domain fluid flow



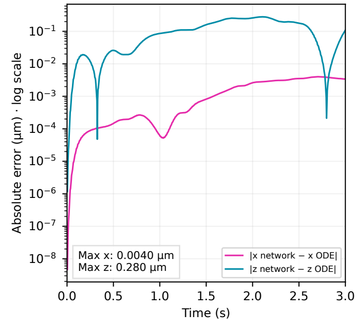

## 04 Network / numerical paths




## 05 Transverse response




## 06 Actual prediction error



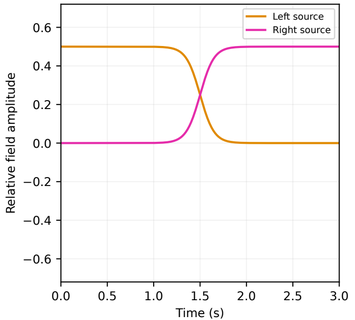

Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Original straight-channel geometry and overdamped equation. No independent launch-velocity input. No-B paths remain straight. Spatial axes use different units.

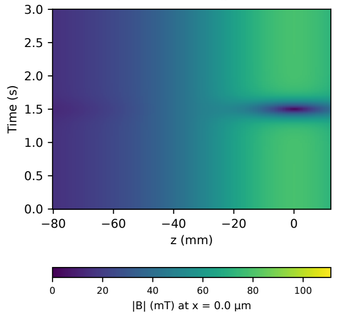
# TRAINED MASTER10 | Smooth switch: left to right

Original case · $x_0$=0.00 µm, $z_0$=-80.0 mm · Umean=15 mm/s, η=3.277 mPa·s · T=3 s

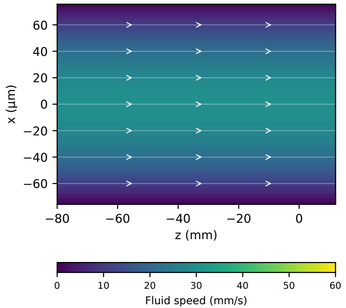
## 01 Magnetic input



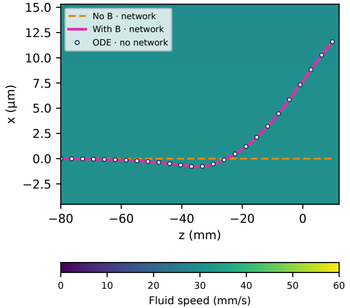

## 02 Space–time magnetic field



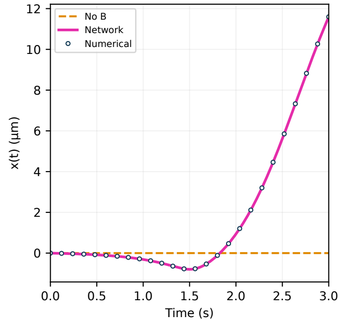

## 03 Trained-domain fluid flow



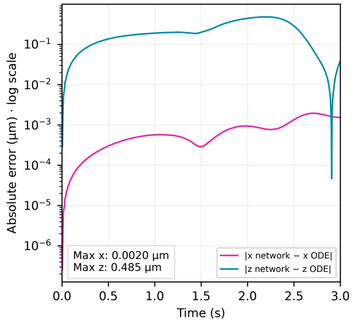

## 04 Network / numerical paths




## 05 Transverse response




## 06 Actual prediction error



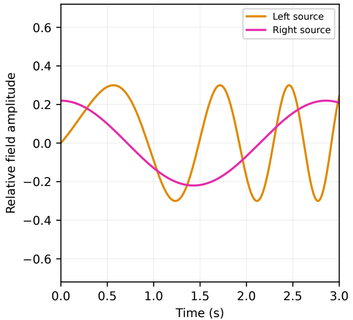

Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Original straight-channel geometry and overdamped equation. No independent launch-velocity input. No-B paths remain straight. Spatial axes use different units.

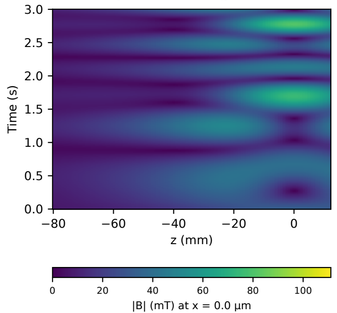
# TRAINED MASTER10 | Two-source chirp and cosine

Original case · $x_0$=0.00 µm, $z_0$=-80.0 mm · Umean=15 mm/s, η=3.277 mPa·s · T=3 s

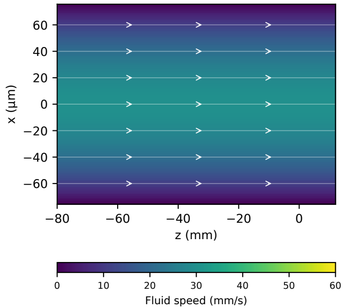
## 01 Magnetic input



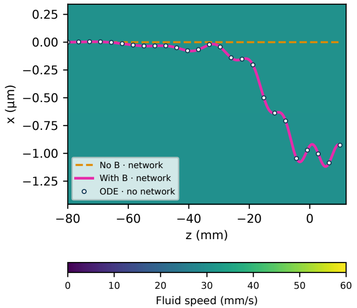

## 02 Space-time magnetic field



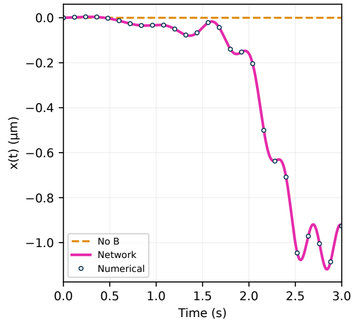

## 03 Trained-domain fluid flow



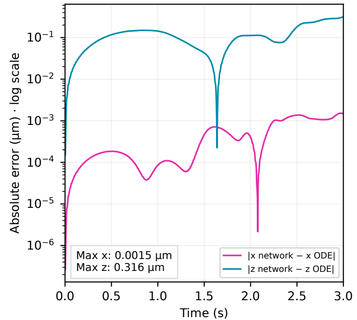

## 04 Network / numerical paths




## 05 Transverse response




## 06 Actual prediction error



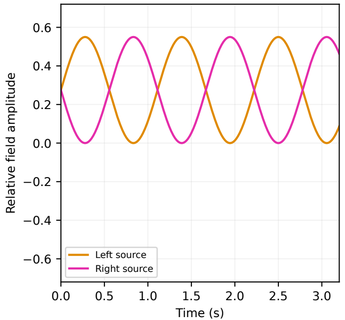

Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Original straight-channel geometry and overdamped equation. No independent launch-velocity input. No-B paths remain straight. Spatial axes use different units.

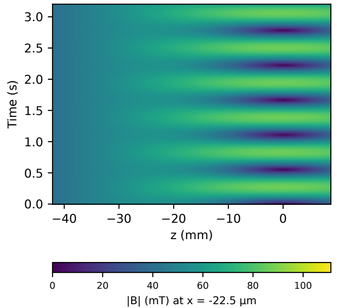
# TRAINED MASTER10 | Off-axis release · alternating attraction

Additional case · $x_0$=-22.50 µm, $z_0$=-40.0 mm · Umean=8 mm/s, η=2.000 mPa·s · T=3.2 s

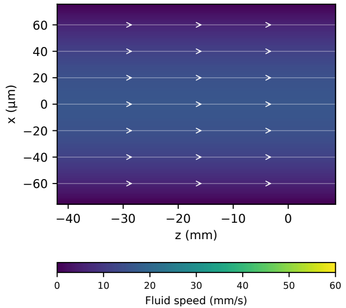
## 01 Magnetic input



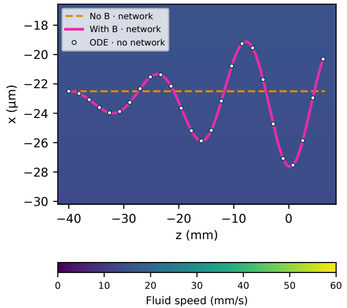

## 02 Space–time magnetic field



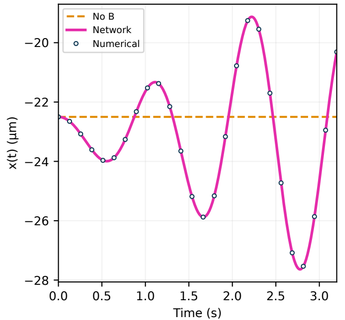

## 03 Trained-domain fluid flow



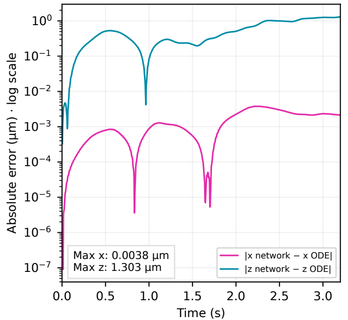

## 04 Network / numerical paths




## 05 Transverse response




## 06 Actual prediction error



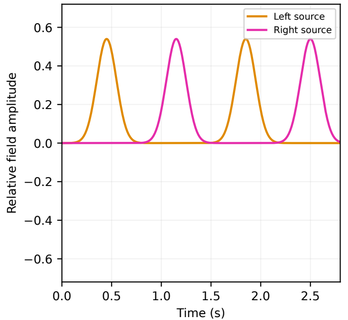

Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Original straight-channel geometry and overdamped equation. No independent launch-velocity input. No-B paths remain straight. Spatial axes use different units.

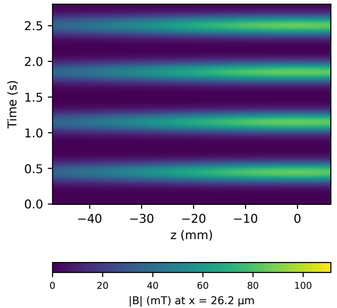
# TRAINED MASTER10 | Shifted release · alternating pulse train

Additional case · $x_0$=26.25 µm, $z_0$=-45.0 mm · Umean=10 mm/s, η=1.500 mPa·s · T=2.8 s

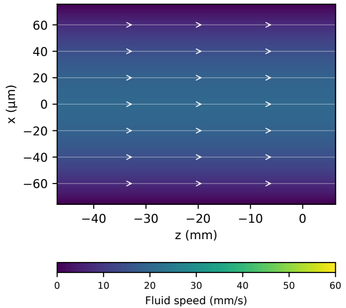
## 01 Magnetic input



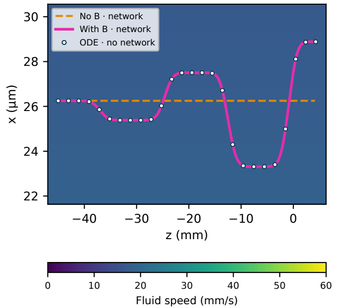

## 02 Space–time magnetic field



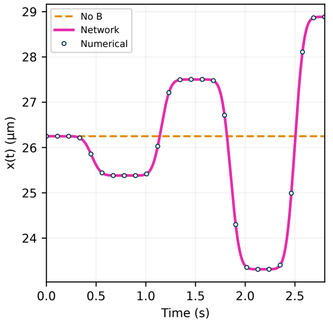

## 03 Trained-domain fluid flow



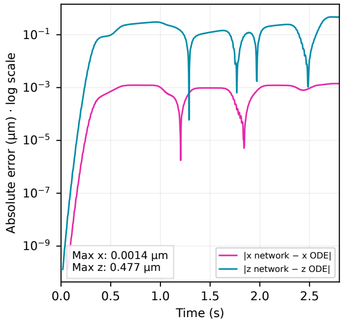

## 04 Network / numerical paths




## 05 Transverse response




## 06 Actual prediction error



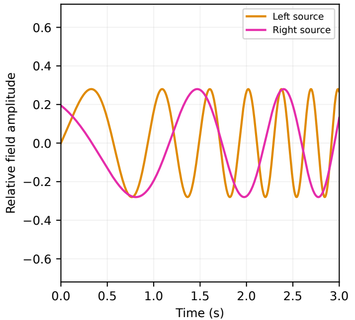

Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Original straight-channel geometry and overdamped equation. No independent launch-velocity input. No-B paths remain straight. Spatial axes use different units.

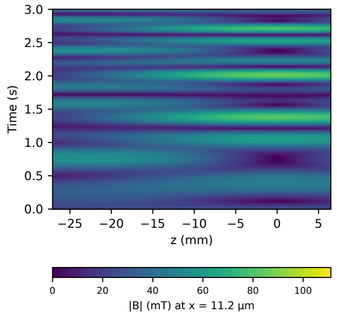
# TRAINED MASTER10 | Slow flow · two-source chirps

Additional case · $x_0$=11.25 µm, $z_0$=-25.0 mm · Umean=5 mm/s, η=1.000 mPa·s · T=3 s

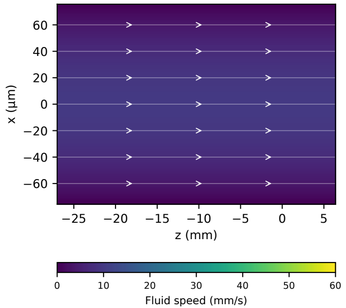
## 01 Magnetic input



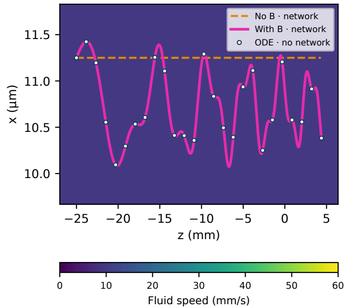

## 02 Space–time magnetic field



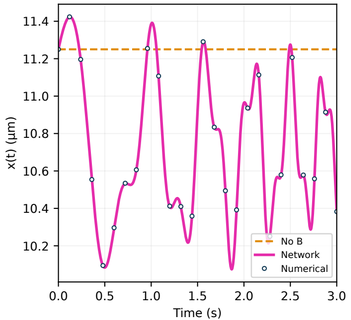

## 03 Trained-domain fluid flow



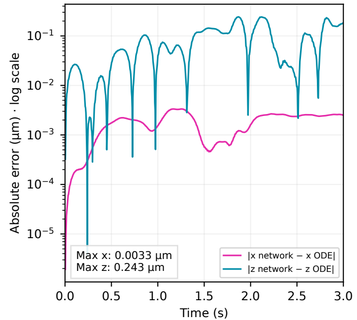

## 04 Network / numerical paths




## 05 Transverse response




## 06 Actual prediction error

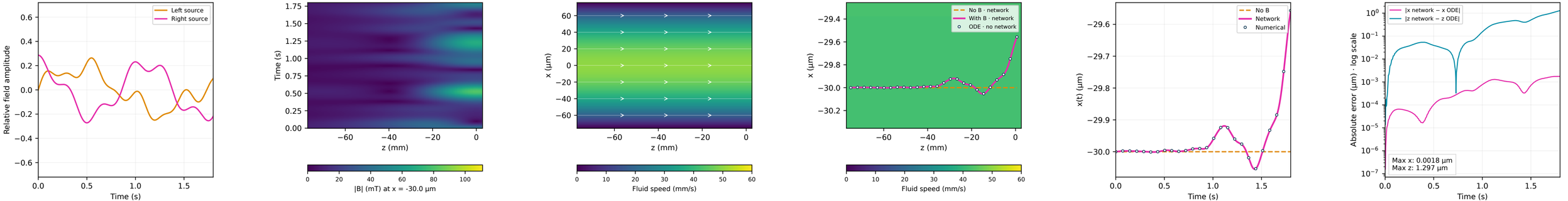



Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Original straight-channel geometry and overdamped equation. No independent launch-velocity input. No-B paths remain straight. Spatial axes use different units.

# TRAINED MASTER10 | Fast flow → multi-frequency excitation

Additional case · $x_0$=-30.00 µm, $z_0$=-75.0 mm · Umean=25 mm/s, η=3.500 mPa·s · T=1.8 s

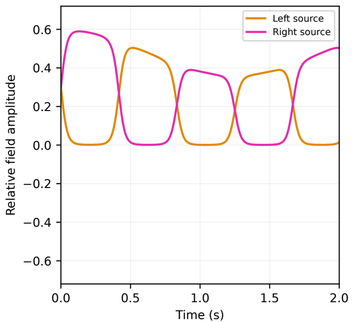


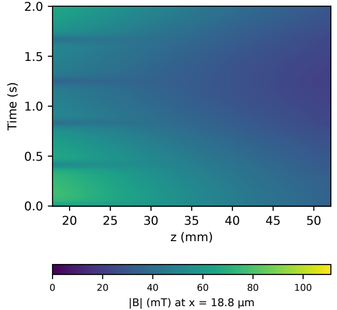

Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

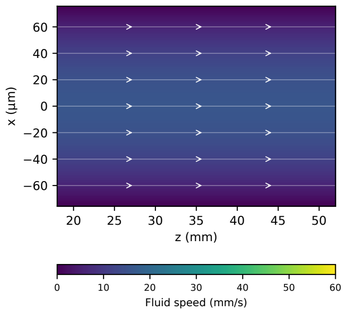
Original straight-channel geometry and overdamped equation. No independent launch-velocity input. No-B paths remain straight. Spatial axes use different units.

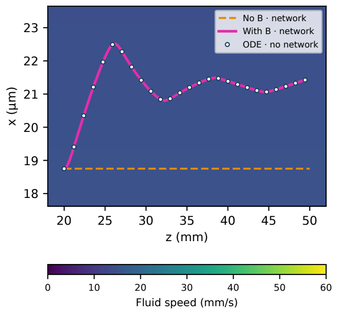
# TRAINED MASTER10 | Downstream release · repeated source switching

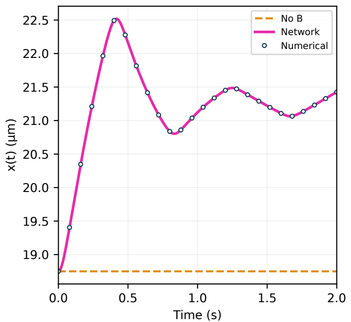
Additional case · $x_0$=18.75 µm, $z_0$=20.0 mm · Umean=8 mm/s, η=2.200 mPa·s · T=2 s

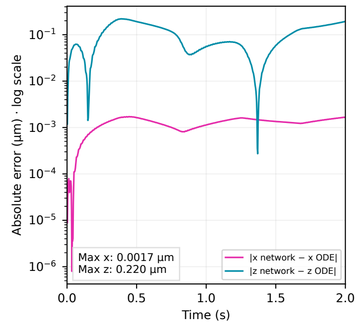


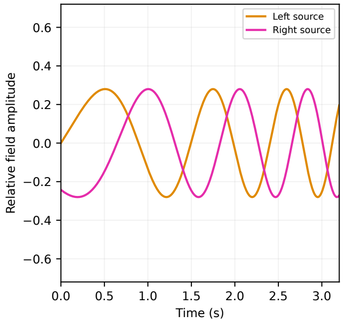

Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Original straight-channel geometry and overdamped equation. No independent launch-velocity input. No-B paths remain straight. Spatial axes use different units.

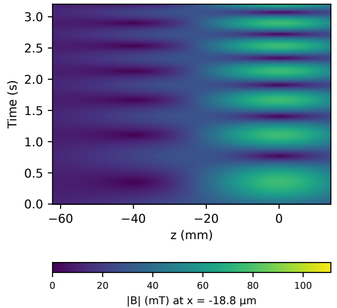
# TRAINED MASTER10 | Shifted release · phase-controlled interference

Additional case · $x_0$=-18.75 µm, $z_0$=-60.0 mm · Umean=12 mm/s, η=4.500 mPa·s · T=3.2 s

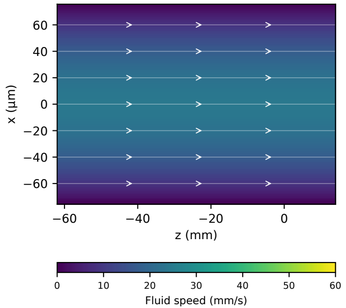
## 01 Magnetic input



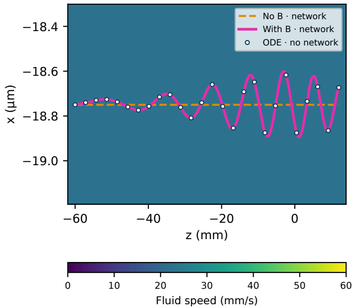

## 02 Space–time magnetic field



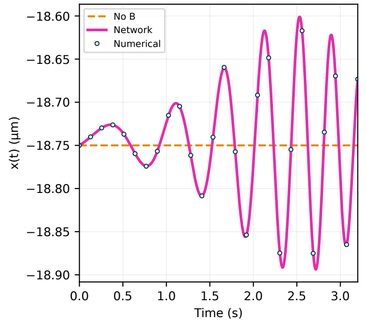

## 03 Trained-domain fluid flow



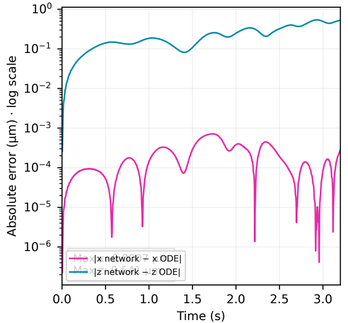

## 04 Network / numerical paths




## 05 Transverse response




## 06 Actual prediction error




Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Original straight-channel geometry and overdamped equation. No independent launch-velocity input. No-B paths remain straight. Spatial axes use different units.